From Classroom/Integrated School Programs 7 in Top 20, 23 in Top 100, 54 in Top 300, 106 in Top 500 All India Ranks & 2314 Students from Classroom /Integrated School Programs & 3723 Students from All Programs have been Awarded a Rank in JEE (Advanced), 2013

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST - I

Paper 2

Time Allotted: 3 Hours **Maximum Marks: 180**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 08)** contains 8 multiple choice questions which have one or more correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (09 to 16) contains 4 paragraphs with each having 2 questions. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

(ii) **Section-A (17 – 20)** contains 4 Matching Lists Type questions: Each question has four statements in LIST I & 4 statements in LIST II. The codes for lists have choices (A), (B), (C), (D) out of which only one is correct. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

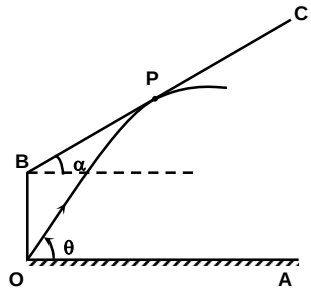
PART – I

SECTION – A

Straight Objective Type

This section contains **8 multiple choice questions** numbered 1 to 8. Each question has 4 choices (A), (B), (C) and (D), out of which only **ONE OR MORE** is/are correct

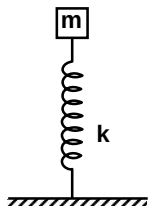
1. A particle is projected from a point O in the horizontal surface OA with speed u and angle of projection θ. It just grazes the plane BC which makes an angle α with the horizontal. The time taken by the projectile to reach p from the instant of projection is



(A) $\frac{2u\sin\theta}{g\cos\alpha}$ (B) $\frac{u\sin\theta}{g\cos\alpha}$

(C) $\frac{u\sin(\theta-\alpha)}{g\cos\alpha}$ (D) None of these.

2. A block of mass m is attached to a spring of force constant k whose other end is fixed to a horizontal surface. Initially the spring is in its natural length and the block is released from rest. The average force acting on the surface by the spring till the instant when the block has zero acceleration for the first time is



(A) $\frac{mg}{\pi}$ (B) $\frac{2mg}{\pi}$

(C) $\frac{3mg}{\pi}$ (D) $\frac{4mg}{\pi}$

3. The gas inside a spherical bubble expands uniformly and slowly so that its radius increases from R to 2R. Let the atmospheric pressure be p_0 and surface tension be S. The work done by the gas in the process is

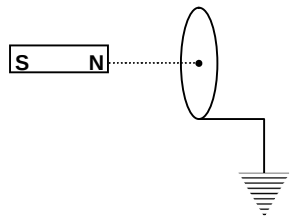
(A) $\frac{28\pi p_0R^3}{3} + 24\pi SR^2$ (B) $\frac{25\pi p_0R^3}{3} + 24\pi SR^2$

(C) $\frac{25\pi p_0R^3}{3} + \frac{23\pi SR^2}{2}$ (D) None of these

Space for Rough work

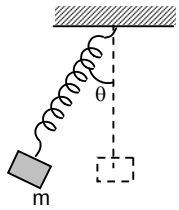
4. A block of mass M is hanging over a smooth and light pulley through a light string. The other end of the string is pulled by a constant force F. If kinetic energy of the block increases by 20 J in 1s. Then
(A) tension in the string is Mg.
(B) tension in the string is F
(C) Work done by the tension on the block is 20 J in 1 sec.
(D) Work done by the force of gravity is 20 J in 1 sec.

5. A circular ring is fixed in a gravity free space and one point of the ring is earthed. Now a magnet is placed along axis of the ring at a distance from its centre such that the nearer pole is north pole as shown in figure. A sharp impulse is applied on the magnet so that it starts to move towards the ring. Then,



(A) Initially magnet experiences an acceleration and then it retards to come to an instantaneous rest.
(B) Magnet starts to oscillate about centre of the ring
(C) Magnet continuous to move along the axis with constant velocity
(D) The magnet retards and comes to rest finally.

6. A block of mass m is suspended by means of an ideal spring of force constant k from ceiling of a car which is moving along a circular path of radius 'r' with acceleration 'a'. The time period of oscillation of the block when it is displaced along the spring, will be



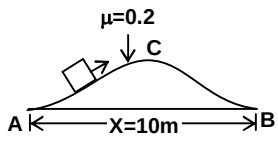
(A) $2\pi\sqrt{\frac{mg+ma}{k}}$ (B) $2\pi\sqrt{\frac{m}{k\sqrt{g^2+a^2}}}$

(C) $2\pi\sqrt{\frac{m}{k}}$ (D) $2\pi\sqrt{\frac{m}{\sqrt{k^2+g^2+a^2}}}$

7. In the hydrogen atom spectrum $\lambda_{3\text{-}1}$ and $\lambda_{2\text{-}1}$ represent wavelengths emitted due to transition from second and first excited states to the ground state respectively. The value of $\frac{\lambda_{3-1}}{\lambda_{2-1}}$ is

(A) 27 / 32 (B) 32/27
(C) 4/9 (D) 9/4

8. A block of mass 1 kg is pulled along the curve path ACB by a tangential force as shown in figure. The work done by the frictional force when the block moves from A to B is



(A) 5 J (B) 10 J
(C) 20 J (D) none of these

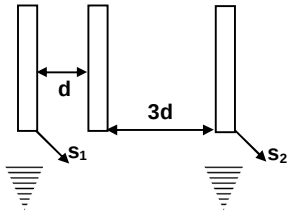
Space for Rough work

Paragraph Type

This section contains 8 **multiple choice questions** relating to **four paragraphs** with **two questions on each paragraph**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

Paragraph for Questions 9 and 10

Three metallic plates out of which middle is given charge Q as shown in the figure. The outer plates can be earthed with the help of switches S_1 and S_2. The area of each plates is same.
Answer the following question based on the following passage.

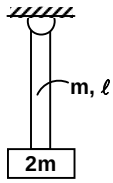


9. The charge appearing on the outer surface of extreme left plate is
(A) $-(Q/2)$ (B) $(Q/2)$
(C) Q (D) $-Q$

10. The charge that will flow to earth when only switch S_1 is connected to earth is
(A) $-(Q/2)$ (B) $(Q/2)$
(C) Q (D) $-Q$

Paragraph for Questions 11 and 12

A block of mass 2m is hanging at the lower end of a rope of mass m and length the ℓ, the other end being fixed to the ceiling. A pulse of wavelength λ_0 is produced at the lower end of the rope.



11. The wavelength of the pulse when it reaches the other end of the rope is
(A) $\sqrt{3}\lambda_0$ (B) $\sqrt{\frac{3}{2}}\lambda_0$
(C) λ_0 (D) $\frac{\lambda_0}{2}$

12. The speed of the pulse at the mid point of rope is
(A) $\sqrt{\frac{5}{2}g\ell}$ (B) $\sqrt{\frac{5}{3}g\ell}$
(C) $\sqrt{\frac{2}{5}g\ell}$ (D) $\sqrt{\frac{g\ell}{2}}$

Space for Rough work

Paragraph for Questions 13 and 14

A bike of length ℓ having its centre of mass at the middle is to turn with constant speed v [here v is the speed of centre of mass of the bike]. To turn, the front wheel is rotated by an angle θ. The coefficient of friction between the tyres and the road is μ. Assuming the bike to be vertical and it does not skid, answer the following questions.

13. The radius of curvature of the centre of mass of the bike is

(A) $\frac{\ell}{\tan\theta}\sqrt{4+\tan^2\theta}$ (B) $\frac{\ell}{\tan\theta}\sqrt{2+\tan^2\theta}$

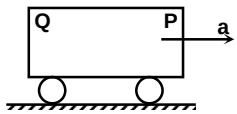
(C) $\frac{\ell}{2\tan\theta}\sqrt{4+\tan^2\theta}$ (D) $\frac{\ell}{2\tan\theta}\sqrt{2+\tan^2\theta}$

14. The friction force acting on the bike

(A) $\frac{2mv^2\tan^2\theta}{\ell\sqrt{4+\tan^2\theta}}$ (B) $\frac{2mv^2\tan^2\theta}{\ell\sqrt{2+\tan^2\theta}}$

(C) $\frac{mv^2\tan^2\theta}{\ell\sqrt{4+\tan^2\theta}}$ (D) $\frac{mv^2\tan^2\theta}{\ell\sqrt{4+\tan^2\theta}}$

Paragraph for Questions 15 and 16



A cylindrical vehicle containing a gas of mass m and molecular weight M moves along the axis of the cylinder with constant acceleration "a". The temperature of the gas is maintained constant at T_0 and saturated vapour pressure at this temperature is P_0. Saturated vapour pressure is the pressure at which condensation starts. The cross sectional area of the cylinder is A and volume V. Now, answer the following questions neglecting gravity and assuming the pressure at the front face to be zero & varies linearly from P to Q.

15. The maximum value of "a" so that no condensation takes place

(A) $\frac{p_0A}{m}$ (B) $\frac{p_0A}{2m}$

(C) $\frac{2p_0A}{m}$ (D) $\frac{3p_0A}{2m}$

16. If $a = 5p_0A/m$, the mass of the gas condensed is

(A) m/2 (B) m/3

(C) 4m/5 (D) m/5

Space for Rough work

Matching list Type

This section contains **4 multiple choice questions**. Each question has matching lists. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

17. Match **List - I** with **List - II** and select the correct answer using the codes given below the lists:

List – I		List – II	
(P)	Isothermal process	(1)	Pressure = constant
(Q)	Adiabatic process	(2)	$dW = -dU$
(R)	Isochoric process	(3)	$dU = 0$
(S)	Isobaric process	(4)	$dQ = dU$

Codes:

	P	Q	R	S
(A)	2	3	1	4
(B)	3	2	4	1
(C)	4	1	2	3
(D)	2	3	4	1

18. Match **List - I** with **List - II** and select the correct answer using the codes given below the lists:

List – I		List – II	
(P)	Ultraviolet light	(1)	$N = 6 \rightarrow n = 3$
(Q)	Visible light	(2)	$N = 3 \rightarrow n = 1$
(R)	Infrared radiation	(3)	$N = 4 \rightarrow n = 2$
(S)	Micro waves	(4)	$N = 7 \rightarrow n = 6$

Codes:

	P	Q	R	S
(A)	2	3	1	4
(B)	1	4	2	3
(C)	4	1	2	3
(D)	2	3	4	1

Space for Rough work

19. Three concentric spherical metallic shells A, B, and C of radii a, b, and c (a<b<c) have charge densities of σ, $-\sigma$ and σ respectively. Match **List - I** with **List - II** and select the correct answer using the codes given below the lists:

List – I		**List – II**	
(P)	The potential at A	(1)	$\frac{\sigma}{\varepsilon_0}\left(\frac{a^2}{c}-\frac{b^2}{c}+c\right)$
(Q)	The potential at B	(2)	$\frac{\sigma}{\varepsilon_0}\left(\frac{a^2}{c}-b+c\right)$
(R)	The potential at C	(3)	$\frac{\sigma}{\varepsilon_0}$
(S)	The electric field at surface of A	(4)	$\frac{\sigma}{\varepsilon_0}(a-b-c)$

Codes:

	P	Q	R	S
(A)	2	3	1	4
(B)	1	4	2	3
(C)	4	1	2	3
(D)	2	3	4	1

20. Match **List - I** with **List - II** and select the correct answer using the codes given below the lists:

List – I		**List – II**	
(P)	Solid sphere is rolling without slipping. The ratio of rotational K.E. to the translational K.E.	(1)	$\frac{5}{7}$
(Q)	When ring and disc both rolling without slipping on the inclined plane the ratio of their times to reach bottom.	(2)	$\frac{2}{5}$
(R)	A solid sphere is rolling with slipping. The ratio of frictional force to normal reaction is (μ = 0.25)	(3)	$\frac{2}{\sqrt{3}}$
(S)	The ratio of translational kinetic energy to the total K.E. when sphere is rolling without slipping on the inclined plane.	(4)	$\frac{1}{4}$

Codes:

	P	Q	R	S
(A)	2	3	1	4
(B)	1	4	2	3
(C)	4	1	2	3
(D)	2	3	4	1

Space for Rough work

Chemistry

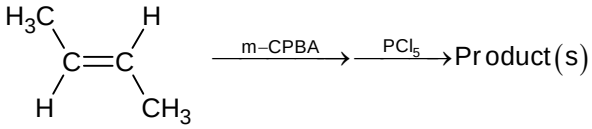
PART – II

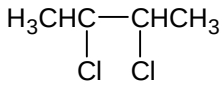
SECTION – A

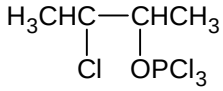
Straight Objective Type

This section contains **8 multiple choice questions** numbered 1 to 8. Each question has 4 choices (A), (B), (C) and (D), out of which only **ONE OR MORE** is/are correct

1. Sometimes yellow turbidity appears while passing H_2S gas even in the absence of IInd group of basic radicals. Which of the following is/are not correct justification for the aforesaid observation?
   (A) Sulphur is present in the mixture as impurity
   (B) IV group radicals are precipitated as sulphides
   (C) H_2S (g) is oxidized by some acid radicals
   (D) IIIrd group radicals are precipitated as hydroxides

2. 

   (A) The product is 

   (B) The product is 

   (C) The product is optically active
   (D) The product is optically inactive

3. Which of the following is / are correct matches?

	Column – I (Trend)		Column – II (Characteristic/Property)
(A)	$NaNO_3 > KNO_3 > RbNO_3$	(p)	Thermal stability
(B)	$MgO > SnO > B_2O_3$	(q)	Basic nature
(C)	$NaHCO_3 < KHCO_3 < RbHCO_3$	(r)	Solubility in H_2O
(D)	$AlF_3 > AlCl_3 > AlBr_3$	(s)	Melting point

4. For the cell:
   $Tl(s) \mid Tl^{+}(0.001\,M) \,||\, Cu^{2+}(0.1\,M) \mid Cu(s)$;
   E_{cell} = 0.83 V at 298 K. The cell potential can be increased by
   (A) increasing $[Cu^{2+}]$ (B) increasing $[Tl^{+}]$
   (C) decreasing $[Cu^{2+}]$ (D) decreasing $[Tl^{+}]$

Space for Rough work

5. Which of the following statement(s) is/are true?
   (A) The equivalent conductance of strong and weak electrolytes increases on dilution.
   (B) The increase in equivalent conductance of strong and weak electrolytes on dilution is due to increase in number of ions.
   (C) If $CH_3COOH(aq)$ is separated from pure water by semipermeable membrane, then equivalent conductance of solution increases gradually and ultimately becomes constant.
   (D) (B) and (C) both are incorrect.

6. 34.1 gm Pb_3O_4 is dissolved in 500 ml of 4M HNO_3, then (M_{Pb} = 206 gm/mole)
   (A) There is no residue
   (B) The weight of residue is 11.9 gm
   (C) 300 ml of 6 M NaOH is required to neutralise excess of HNO_3
   (D) Total 2.2 moles of NaOH are used after reaction of Pb_3O_4 with HNO_3

7. Which of the following statement(s) is/are correct?
   (A) During electrophoresis, colloidal particles move under influence of electric field.
   (B) Suspensions are opaque and they do not show Tyndell effect.
   (C) Higher is the coagulating power of an electrolyte, higher is its coagulation value.
   (D) 50 ml of standard gold sol needs 0.05 mg of gelation or 125 mg of starch, for its protection from coagulation by 1 ml, 10% NaCl solution, it means gold number of starch is more than that of gelation and is better protective agent than gelation.

8. A balloon blown up with 1 mole of a gas, has a volume of 480 ml to 5^oC. If the balloon is filled to $\left(\frac{7}{8}\right)$th of its maximum capacity, then which of the following options is/are correct?
   (A) The balloon will burst at 30^oC
   (B) The pressure of the gas inside the balloon at 5^oC is 47.5 atm
   (C) The minimum temperature at which the balloon will burst is 44.71^oC
   (D) The pressure of gas when balloon burst at minimum temperature is 50 atm

Space for Rough work

Paragraph Type

This section contains 8 **multiple choice questions** relating to **four paragraphs** with **two questions on each paragraph**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

Paragraph for Questions 9 and 10

Answer the questions followed by the given reaction sequence

$$\underset{(\text{white powder})}{A} \xrightarrow{\Delta} \underset{\text{odorless colorless}}{B(g)} + C(s)$$

$$B(g) + Ca(OH)_2 \longrightarrow \underset{\text{milky white}}{D\downarrow} + H_2O$$

$$D(s) + B(g) \xrightarrow{H_2O} E$$

$$B(g) + K_2Cr_2O_7 \longrightarrow \text{No response}$$

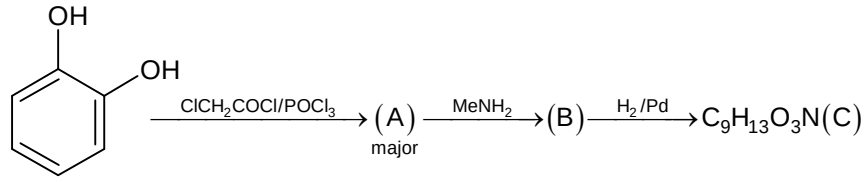
C(s) is white when cold and turns yellow when heated.

9. What is A?
   (A) Li_2CO_3 (B) Mg_2CO_3
   (C) Na_2CO_3 (D) $ZnCO_3$

10. C and E are respectively:
   (A) MgO, $Ca(HCO_3)_2$ (B) Li_2O, $Ca(HCO_3)_2$
   (C) ZnO, $Ca(HCO_3)_2$ (D) ZnO, $Mg(HCO_3)_2$

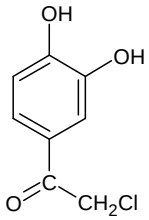
Space for Rough work

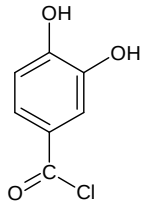
Paragraph for Question Nos. 11 to 12

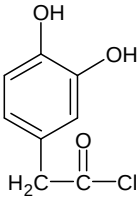


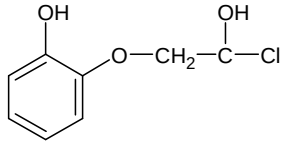
'A' also reacts with NaOI followed by the reaction with H^+ to give 3,4 – dihdyroxybenzoic acid.

11. What is A?

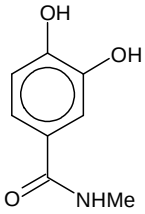
(A) 

(B) 

(C) 

(D) 

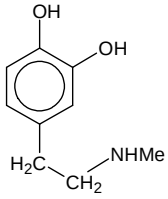
12. Which of the following is correct?



(A) 'B' is

(B) 'C' is optically active which is derived from A

(C) no other side product is formed along with 'A'

(D) 'C' is 

Space for Rough work

Paragraph for Question Nos. 13 to 14

An organic compound (A) $C_3H_6Cl_2$ on reduction with red P and HI gives propane. (A) on hydrolysis by an alkali followed by oxidation gives (B) $C_3H_4O_4$, which on heating gives (C). Both (B) and (C) give effervescence with $NaHCO_3$. (B) on reacting with ethyl alcohol gives (D) $C_7H_{12}O_4$, a well known synthetic reagent.

Now answer the following questions

13. $(D) + \text{Benzaldehyde} \xrightarrow[(ii)\ H_3O^+]{(i)\ \text{Pyridine},\ \Delta} \text{Major Prodcut}(E)$

The major product (E) is

(A) crotonic acid
(B) cinnamic acid
(C) benzoic acid
(D) Mandelic acid

14. Compound (C) $\xrightarrow{P_4O_{10}}$ Product (F)

Hence product F may be

(A) malonic anhydride
(B) ethanoic anhydride
(C) ethanal
(D) a lactone

Space for Rough work

Paragraph for Question Nos. 15 to 16

The statement of the first law of thermodynamics given by Clausius, is: 'Die Energie der welt ist konstant'. That is, the energy of the universe is constant.
Mathematically it can be written as:

$\sum \Delta U = 0; \quad (\text{universe})$

The first law predicts that energy added to or removed from a system must be accounted for, by a change in the internal energy ΔU. Energy is added to or removed from a system as work (w) or heat (q), in which case $\Delta U = q + w$;

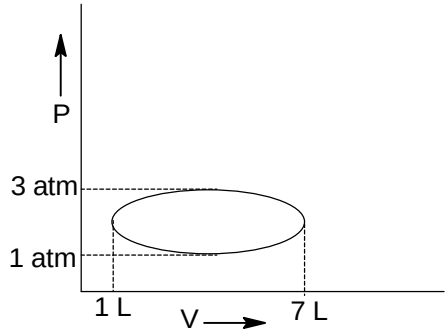
for an infinitesimal process, this equation becomes $\delta U = \delta q + \delta w$

Heat and work represent ways that energy can be transferred from one system to another. q and w, both are path functions and both are interconvertible. Extent of work done can be calculated by

$w = -\int_{V_i}^{V_f} P_{ext} dV.$

15. Which of the following is correct statement?
    (A) The sum of the two path dependent thermodynamic properties, is always path dependent.
    (B) Work appears at the boundary of the system as well as inside the system.
    (C) Work and heat are completely interconvertible in real life systems.
    (D) None of the above.

16. Which of the following is correct?
    (A) Work done in a cyclic process is always zero.
    (B) A two litre container is filled with the gas at 10^{-3} atm pressure and the gas is allowed to diffuse into an empty and closed container of volume 4 litre. Then work done in the process is zero due to no opposing force.
    (C) PV work done in a cyclic process, which is represented as follows (for 1 mole ideal gas), 953 Joule



    (D) (B) and (C) both are correct.

Space for Rough work

Matching list Type

This section contains **4 multiple choice questions**. Each question has matching lists. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

17. Match the following column:

List – I		List – II	
(P)	$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$	(1)	Increase in volume favours forward reaction
(Q)	$2NO(g) \rightleftharpoons N_2(g) + O_2(g)$	(2)	Increase in pressure accelerate the forward reaction
(R)	$CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$	(3)	$K_p = P_{total}$
(S)	$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$	(4)	At constant volume introduction of inert gas, has no effect.

Codes:

	P	Q	R	S
(A)	2	4	3	1
(B)	2	3	1	4
(C)	3	1	4	2
(D)	4	3	1	2

18. Match the following

List – I (Species)		List – II (Hybridisation of central atom)	
(P)	$(BN)_x$	(1)	sp^3
(Q)	$(BeCl_2)_n$; (n > 2)	(2)	sp^2
(R)	$[Cu(NH_3)_4]^{2+}$	(3)	dsp^2
(S)	MnO_4^-	(4)	d^3s

Codes:

	P	Q	R	S
(A)	3	4	1	2
(B)	2	1	3	4
(C)	3	1	4	2
(D)	4	3	1	2

Space for Rough work

19. Match the following

List – I (Thermodynamics process)	List – II (ΔS_{System})
(P) Isothermal reversible	(1) $nc_v \ln\frac{T_2}{T_1}$
(Q) Adiabatic irreversible	(2) $nc_p \ln\frac{T_2}{T_1}$
(R) Isochoric reversible	(3) $nc_v \ln\frac{T_2}{T_1} + nR\ln\frac{V_2}{V_1}$
(S) Isobaric reversible	(4) $nR\ln\frac{V_2}{V_1}$

Codes:

	P	Q	R	S
(A)	3	4	1	2
(B)	2	1	3	4
(C)	3	1	4	2
(D)	4	3	1	2

20. Match the following

List – I (Species)	List – II (relation)
(P) CN^-, N_2	(1) Isodiaphers
(Q) F^{19}, K^{39}	(2) Isobars
(R) N^{15}, O^{15}	(3) Isotones
(S) C^{14}, N^{15}	(4) Isoesters

Codes:

	P	Q	R	S
(A)	3	4	1	2
(B)	2	1	3	4
(C)	3	1	4	2
(D)	4	1	2	3

Space for Rough work

Mathematics

PART – III

SECTION – A

Straight Objective Type

This section contains **8 multiple choice questions** numbered 1 to 8. Each question has 4 choices (A), (B), (C) and (D), out of which only **ONE OR MORE** is/are correct

1. A non-zero vector $\vec{a}$ is parallel to the line of intersection of the plane P_1 determined by $\hat{i}+\hat{j}$ and $\hat{i}-2\hat{j}$ and plane P_2 determined by vector $2\hat{i}+\hat{j}$ and $3\hat{i}+2\hat{k}$, then angle between $\vec{a}$ and vector $\hat{i}-2\hat{j}+2\hat{k}$ is

   (A) $\frac{\pi}{4}$ (B) $\frac{\pi}{2}$

   (C) $\frac{\pi}{3}$ (D) none of these

2. Let f(x) be a function satisfying f′(x) = f(x) with f(0) = 1 and g be the function satisfying f(x) + g(x) = x, then the value of $\int_0^1 f(x)g(x)dx$

   (A) $\frac{3-e^2}{2}$ (B) $\frac{e^2-3}{2}$

   (C) $\frac{e^2}{2}$ (D) $\frac{e-2}{4}$

3. If three one by one squares are selected at random from the chessboard, then the probability that they form the letter 'L' is

   (A) $\frac{196}{{}^{64}C_3}$ (B) $\frac{49}{{}^{64}C_3}$

   (C) $\frac{36}{{}^{64}C_3}$ (D) $\frac{98}{{}^{64}C_3}$

Space for rough work

4. If $x^2 + 9y^2 + 25z^2 = xyz\left(\frac{15}{x}+\frac{5}{y}+\frac{3}{z}\right)$, the x, y, z are in

(A) A.P. (B) G.P.
(C) A.G.P. (D) H.P.

5. The number of ways of choosing triplet (x, y, z) such that $z \geq \max\{x, y\}$ and $x, y, z \in \{1, 2, \ldots, n, n + 1\}$ is

(A) ${}^{n+1}C_3 + {}^{n+2}C_3$ (B) $\frac{1}{6}n(n + 1)(2n + 1)$
(C) $1^2 + 2^2 + \ldots + n^2$ (D) $2({}^{n+2}C_3) - {}^{n+1}C_2$

6. A complex number z is rotated in anticlockwise direction by an angle α and we get z′ and if the same complex number z is rotated by an angle α in clockwise direction and we get z″ then
(A) z′, z, z″ are in G.P. (B) $z'^2 + z''^2 = 2z^2 \cos 2\alpha$
(C) $z' + z'' = 2z \cos \alpha$ (D) z′, z, z″ are in H.P.

7. The equation of the circle which touches the axes of coordinates and the line $\frac{x}{3}+\frac{y}{4}=1$ and whose centre lies in the first quadrant is $x^2 + y^2 - 2cx - 2cy + c^2 = 0$ where c is
(A) 1 (B) 2
(C) 3 (D) 6

8. If $\cos\alpha = \frac{1}{2}\left(x+\frac{1}{x}\right)$ and $\cos\beta = \frac{1}{2}\left(y+\frac{1}{y}\right)$, $(xy > 0)$ x, y, $\alpha, \beta \in R$ then
(A) $\sin(\alpha + \beta + \gamma) = \sin\gamma \ \forall \ \gamma \in R$ (B) $\cos\alpha \cos\beta = 1 \ \forall \ \alpha, \beta \in R$
(C) $(\cos\alpha + \cos\beta)^2 = 4 \ \forall \ \alpha, \beta \in R$ (D) $\sin(\alpha + \beta + \gamma) = \sin\alpha + \sin\beta + \sin\gamma \ \forall \ \alpha, \beta, \gamma \in R$

Space for rough work

Paragraph Type

This section contains 8 **multiple choice questions** relating to **four paragraphs** with **two questions on each paragraph**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

Paragraph for Questions 9 and 10

Read the following write up carefully and answer the following questions:
In an argand plane z_1, z_2 and z_3 are respectively the vertices of an isosceles triangle ABC with AC = BC and $\angle CAB = \theta$. If z_4 is incentre of triangle. Then

9. The value of $\left(\frac{AB}{IA}\right)^2\left(\frac{AC}{AB}\right)$

(A) $\frac{(z_2-z_1)(z_1-z_3)}{(z_4-z_1)^2}$ (B) $\frac{(z_2-z_1)(z_3-z_1)}{(z_4-z_1)}$

(C) $\frac{(z_2-z_1)(z_3-z_1)}{(z_4-z_1)^2}$ (D) none of these

10. The value of $(z_4-z_1)^2$ $(1+\cos\theta)\sec\theta$ is

(A) $(z_2-z_1)(z_3-z_1)$ (B) $\frac{(z_2-z_1)(z_3-z_1)}{(z_4-z_1)}$

(C) $\frac{(z_2-z_1)(z_3-z_1)}{(z_4-z_1)^2}$ (D) $(z_2-z_1)(z_3-z_1)^2$

Space for rough work

Paragraph for Question Nos. 11 to 12

Read the following write up carefully and answer the following questions:

If the normal to the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ at any point P (a cos θ, b tan θ) meets the transverse and conjugate axes in G and g respectively and if 'f' is the foot of perpendicular to the normal at P from the centre 'C' then

11. The length of PG^2 is

(A) $\frac{b^2}{a^2}(b^2\text{cosec}^2\theta+a^2\cot^2\theta)$ (B) $\frac{a^2}{b^2}(b^2\text{cosec}^2\theta+a^2\cot^2\theta)$

(C) $\frac{b^2}{a^2}(b^2\sec^2\theta+a^2\tan^2\theta)$ (D) $\frac{b^2}{a^2}(b^2\tan^2\theta+a^2\sec^2\theta)$

12. Minimum of PG is

(A) $\frac{b^2}{a}$ (B) $\left|\frac{a}{b}(a+b)\right|$

(C) $\left|\frac{b}{a}(a-b)\right|$ (D) $\left|\frac{a}{b}(a-b)\right|$

Paragraph for Question Nos. 13 to 14

Read the following write up carefully and answer the following questions:
If a cricket team of 11 players is to be selected from 8 batsman, 6 bowlers, 4 all rounder and 2 wicket keepers, then

13. The number of selections when atmost 1 all rounder and 1 wicket keeper will play, is
(A) ${}^4C_1\cdot{}^{14}C_{10}+{}^2C_1\cdot{}^{14}C_{10}+{}^4C_1\cdot{}^2C_1\,{}^2C_1\cdot{}^{14}C_9+{}^{14}C_{11}$
(B) ${}^4C_1\cdot{}^{15}C_{11}+{}^{15}C_{11}$
(C) ${}^4C_1\cdot{}^{15}C_{10}+{}^{15}C_{11}$
(D) none of these

14. Number of selection when 2 particular batsmen don't want to play when a particular bowler will play, is
(A) ${}^{17}C_{10}+{}^{19}C_{11}$ (B) ${}^{17}C_{10}+{}^{19}C_{11}+{}^{17}C_{11}$
(C) ${}^{17}C_{10}+{}^{20}C_{11}$ (D) ${}^{19}C_{10}+{}^{19}C_{11}$

Space for rough work

Paragraph for Question Nos. 15 to 16

Read the following write up carefully and answer the following questions:
A curve y = f(x) passes through the point P(1, 1). The normal to the curve at P is a(y – 1) + (x – 1) = 0. If the slope of the tangent at any point on the curve is proportional to the ordinates of the point, then

15. The equation of the curve y = f(x) is
(A) $y = e^{a(x-1)}$ (B) $y = e^{a(x^2-1)}$
(C) $y = e^{a(1-|x|)}$ (D) $y = e^{a(|x|-1)}$

16. The area bounded by the curve y = f(x), y-axis and the normal to the curve at P is
(A) $\frac{1}{a}\left(a-\frac{1}{2}+e^{-a}\right)$ (B) $\frac{1}{a}\left(a+\frac{1}{2}+e^{-a}\right)$
(C) $\frac{1}{a}\left(a-\frac{1}{2}+e^{a}\right)$ (D) $\frac{1}{a}\left(a+\frac{1}{2}+e^{a}\right)$

Matching list Type

This section contains **4 multiple choice questions**. Each question has matching lists. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

17. Match the following List–I with List–II.

	List – I		List – II
(P)	The value of k for which $\lim_{x\to1}\operatorname{cosec}^{-1}\left(\frac{k^2}{\ln x}-\frac{k^2}{x-1}\right)$ exists is	1.	$\left(0, \frac{2}{3}\right)$
(Q)	The value of k for which $kx^2 - 2kx + 3x - 6$ is positive for exactly two integral values of x is	2.	[–1, 4]
(R)	The value of k for which the point (2k + 1, k – 1) is an interior point of the smaller segment of the circle $x^2 + y^2 – 2x – 4y – 4 = 0$ w.r.t. the chord x + y – 2 = 0 is	3.	$\left(-\frac{3}{4}, -\frac{3}{5}\right)$
(S)	The solution of the inequality $\log_{1/5}(2x + 5) + \log_5(16 – x^2) \le 1$ is	4.	$(-\infty, -\sqrt{2}) \cup (\sqrt{2}, \infty)$

Codes:

	P	Q	R	S
(A)	4	3	2	1
(B)	2	1	4	3
(C)	4	3	1	2
(D)	2	4	3	1

Space for rough work

18. If $\int_0^\infty e^{-x^2}dx = \frac{\sqrt{\pi}}{2}$, then match the following List–I with List–II.

List – I		List – II	
(P)	$\int_0^\infty e^{-2x^2}dx$	1.	$\sqrt{\pi}$
(Q)	$\frac{d}{dt}\int_0^\infty e^{-tx^2}dx$ at t = 1	2.	$\frac{3\sqrt{\pi}}{8}$
(R)	$\frac{d^2}{dt^2}\int_0^\infty e^{-tx^2}dx$ at t = 1	3.	$-\frac{\sqrt{\pi}}{4}$
(S)	$\int_{-\infty}^\infty e^{-x^2}dx$	4.	$\frac{\sqrt{\pi}}{2\sqrt{2}}$

Codes:

	P	Q	R	S
(A)	4	3	1	2
(B)	3	4	1	2
(C)	2	1	2	4
(D)	4	3	2	1

19. If in ΔABC, r = 2, r_1 = 4, s = 12 and a < b < c, then match the following List–I with List–II.

List – I		List – II	
(P)	Area of triangle is	1.	5
(Q)	R is equal to	2.	24
(R)	$\angle A$ of triangle is	3.	$\sin^{-1}\frac{4}{5}$
(S)	$\angle B$ of triangle is	4.	$\sin^{-1}\frac{3}{5}$

Codes:

	P	Q	R	S
(A)	4	2	3	1
(B)	2	1	4	3
(C)	3	4	1	2
(D)	2	1	3	4

Space for rough work

20. Match the following List–I with List–II.

	List – I		List – II
(P)	The normal chord at a point t on the parabola $y^2 = 4x$ subtends a right angle at the vertex, then t^2 is	1.	4
(Q)	The area of the triangle inscribed in the curve $y^2 = 4x$, the parameter of coordinates whose vertices are 1, 2 and 4 is	2.	2
(R)	The number of distinct normal possible from $\left(\frac{11}{4}, \frac{1}{4}\right)$ to the parabola $y^2 = 4x$ is	3.	3
(S)	The normal at (a, 2a) on $y^2 = 4ax$ meets the curve again at $(at^2, 2at)$, then the value of $\|t - 1\|$ is	4.	6

Codes:

	P	Q	R	S
(A)	2	4	1	3
(B)	3	1	4	2
(C)	1	3	2	4
(D)	2	4	3	1

Space for rough work